Climate-Risk News: A Useful Signal for Agricultural Commodities?

Replication, out-of-sample prediction, trading, and machine learning

ENSAE Paris | Machine Learning for Portfolio Management and Trading

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Research question and contribution

Can climate-risk news forecast agricultural returns, and is the signal tradable?

1. Replicate

Predictive regressions based on Ma, Zhou & Li (2024), with updated news indicators and a 2005–2025 sample.

2. Make it tradable

Move from commodity indices to agricultural futures and DBA; include real-time estimation and turnover costs.

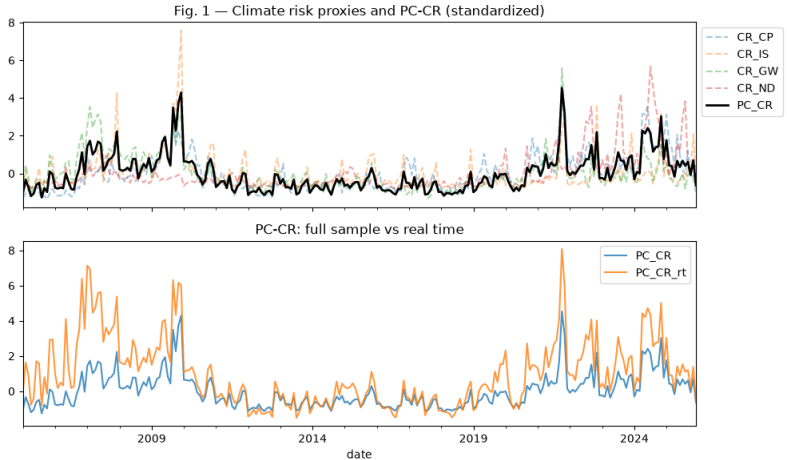
3. Extend the signal

Add Google search attention and compare linear forecasts with four machine-learning models.

Data in brief

Reuters-based climate news (4 themes), World Bank price indices, 7 agricultural futures, DBA, FRED/macroeconomic controls, and Google Trends; monthly frequency.

Constructing the climate-risk signal



Source: notebook cell 11. First principal component of four news-based proxies.

Four news topics

US policy, international summits, global warming, natural disasters.

Two aggregate indices

Full-sample PC-CR for replication; recursively estimated PC-CR for out-of-sample tests.

Global warming emerges as the strongest single predictor for the futures basket.

Empirical design: forecasting without future returns

$$\underbrace{\overline{R}_{t+1:t+h}}_{\text{mean monthly return over next } h \text{ months}} = \alpha + \beta\,\mathrm{ClimateRisk}_t + \varepsilon_{t+h}, \qquad h \in \{1, 6, 12\}.$$

Statistical inference

Newey–West and Hodrick statistics for overlapping forecast horizons.

Out-of-sample evaluation

Recursive re-estimation, realized training targets only, and the historical mean as the benchmark.

Forecast comparison

$$R^2_{\mathrm{OS}} = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \bar{y}_t)^2}.$$

Positive values indicate a lower squared forecast error than the prevailing mean.

Clark–West statistics are used for the linear nested forecast comparison.

Initial estimation: 2005–2012	Validation: 2013–2018	Evaluation: 2019–2025

Climate news predicts returns out of sample

Asset	Predictor	R^2_{OS} 1m	R^2_{OS} 6m	R^2_{OS} 12m	CW 6m
Futures basket	Global warming	1.28	**12.74**	**13.43**	2.73
Futures basket	Real-time PC-CR	0.47	11.41	9.99	2.25
DBA	Global warming	1.05	8.55	7.06	2.43
DBA	Real-time PC-CR	1.33	**11.46**	**10.88**	2.43

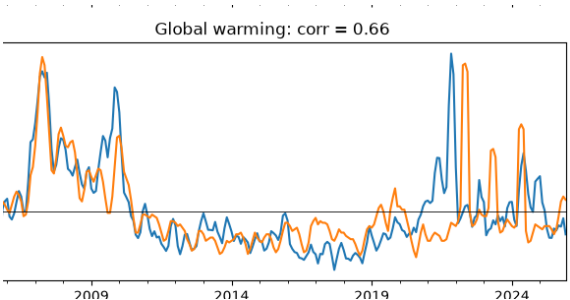
R^2_{OS} figures in %; CW statistics are one-sided. All displayed six-month CW statistics exceed 1.96.

The signal is strongest at 6–12 months. Global-warming news is more predictive for the futures basket; PC-CR is competitive for DBA.

Predictive effects also survive many macroeconomic controls, especially for the futures basket.

Source: notebook cells 32, 35 and 39. Results are predictive, not causal.

Does investor attention add predictive information?



News and searches for global warming; notebook cell 19.

Correlation: 0.66

News and Google search activity capture related but different information.

Six-month R^2_{OS} (%)

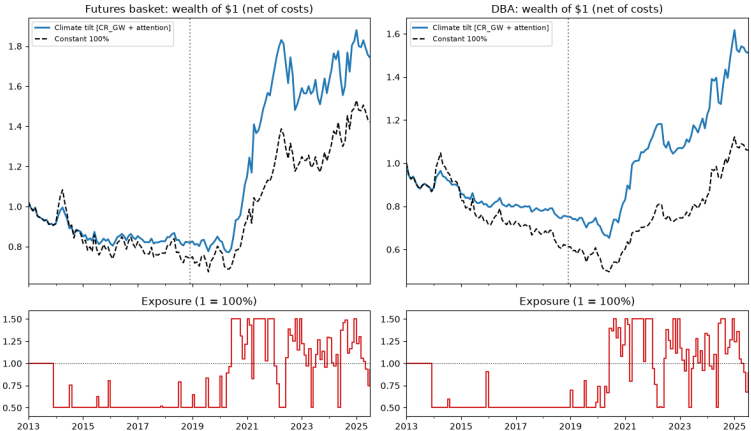
Futures: **12.74** (news) → **13.34** (+ attention)

DBA: **8.55** (news) → **11.45** (+ attention)

Attention gives a modest improvement at six months; the gains are not systematic across horizons.

Source: notebook cells 19 and 48. Abnormal searches are measured against their previous 12 months.

From a forecast to a trading rule



Dynamic exposure

$w_{t+1} = \mathrm{clip}_{[0.5,1.5]}(1 + 0.5 s_t)$

s_t: forecast minus historical mean, divided by its expanding standard deviation.

Futures basket

Sharpe: 0.28 versus 0.16 (constant exposure).

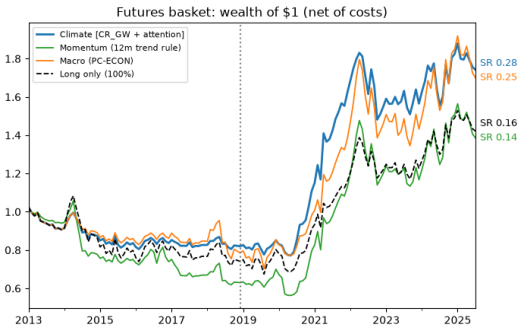
DBA

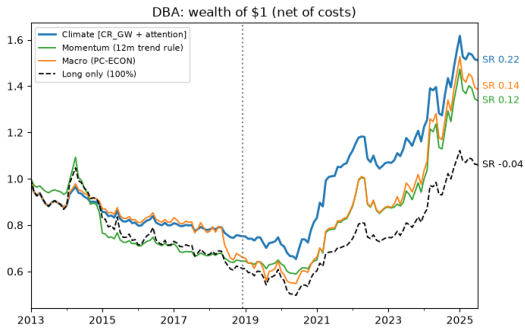
Sharpe: 0.22 versus -0.04.

Net of 30 bp per unit of turnover.

Source: notebook cells 52–54. Selected predictor: global warming + attention. Full OOS period.

Is the strategy better than simple alternatives?





Benchmarks: 12-month trend-following momentum, a macroeconomic forecast, and constant 100% exposure.

Result: the climate tilt has the highest Sharpe ratio on both assets in this comparison (0.28 futures; 0.22 DBA).

The climate tilt compares favorably, mainly by limiting drawdowns.

Source: notebook cell 72. Common evaluation dates; net of assumed trading costs.

Can more flexible machine-learning models do better?

All models use recursive out-of-sample estimation and the same target: average future monthly basket returns.

Model (6-month horizon)	R^2_{OS} vs mean (%)	R^2_{OS} vs linear GW (%)
Linear global-warming risk	**12.74**	0.00
Selected strategy predictor	**13.34**	0.69
Elastic Net	−5.21	−20.49
Random Forest	−12.74	−29.11
Gradient Boosting	−41.04	−61.52
Shallow 1D CNN	−149.41	−188.18

Extra model flexibility does not improve predictive accuracy in this relatively small monthly dataset.

This is consistent with overfitting or unstable relationships; it does not prove the data-generating process is linear.

Source: notebook cell 69, 6-month output. Recalculate all models against a common mean benchmark.

Takeaways and limitations

What the evidence supports

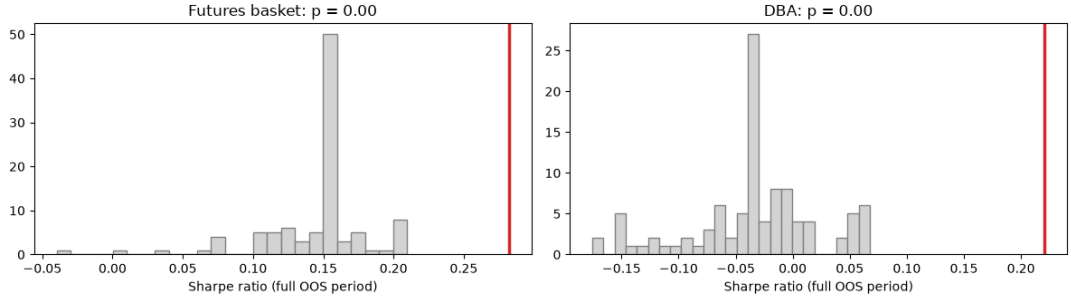
1. Climate-risk news predicts agricultural returns, mainly at 6–12 months.
2. A climate-based exposure tilt improves Sharpe ratios in the backtest, after assumed costs.
3. Search attention adds limited information; complex ML models underperform simple linear forecasts.

What prevents a stronger trading claim

- **Validation:** six-month targets through December 2018 overlap the January 2019 test. The selection cutoff needs to move back six months.
- **Implementation:** continuous futures rolls and Google Trends vintages limit strict real-time tradeability.
- **Sample dependence:** a few persistent exposure regimes drive part of the returns.

Promising predictive evidence; investable performance needs stricter validation.

Appendix — Placebo test: is the signal better than shuffled predictors?



Procedure: randomly circularly shift the candidate signals by at least two years; repeat predictor selection and compute the resulting Sharpe ratio.

Result: none of 100 shifts exceeded the observed Sharpe for either asset. This is exploratory robustness evidence, not a literal zero p-value.

Source: notebook cell 62.